Triangle & Channel Detection Fix — Results

Pivot-Point Fitting + Theil-Sen Regression + Containment Validation

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Summary. The triangle and channel detectors were rebuilt using pivot-point pre-selection (swing highs/lows), Theil-Sen robust regression, and a hard containment validation gate. This document presents the before-vs-after comparison.

1. Before vs After — Detection Counts

Metric	Before (OLS)	After (Pivot + Theil-Sen)
Triangle detections	38	6
ascending_triangle	3	3
descending_triangle	29	0
desc_triangle_upper_test	6	3
symmetric_triangle	0	0
Channel detections	43	1
channel_up	43	0
channel_down	0	1
Total pattern detections	81	7

The large reduction (81 → 7) is intentional. The old detector fired on formations that did not visually match their claimed pattern type. The surviving 7 detections pass strict quality checks and all have trendlines that actually contain the price action.

2. Containment Ratio Comparison

Containment ratio = fraction of bars where High ≤ upper trendline + tol AND Low ≥ lower trendline – tol (tolerance = 0.1 × ATR).

	Before (OLS)	After (Pivot + Theil-Sen)
Triangle mean containment	3.8%	87.7%
Triangle min containment	0.0%	82.0%
Channel mean containment	8.5%	74.0%
Channel min containment	0.0%	74.0%
Threshold enforced	None	80% (triangles), 70% (channels)

Key result: containment improved from ~4–9% (OLS) to 74–96% (pivot + Theil-Sen). Every detection now has trendlines that visibly bound the price action — directly addressing the supervisor's feedback.

3. What Changed in the Code

3.1 New shared module: src/patterns/pivots.py

- **find_swing_highs(highs, order=5)** — identifies bars where High is the maximum in a ±5-bar window (confirmed local peaks).
- **find_swing_lows(lows, order=5)** — same for local troughs.
- **containment_ratio(highs, lows, upper, lower, tol)** — fraction of bars fully inside the pattern boundaries.

3.2 Triangle detector (src/patterns/triangles.py)

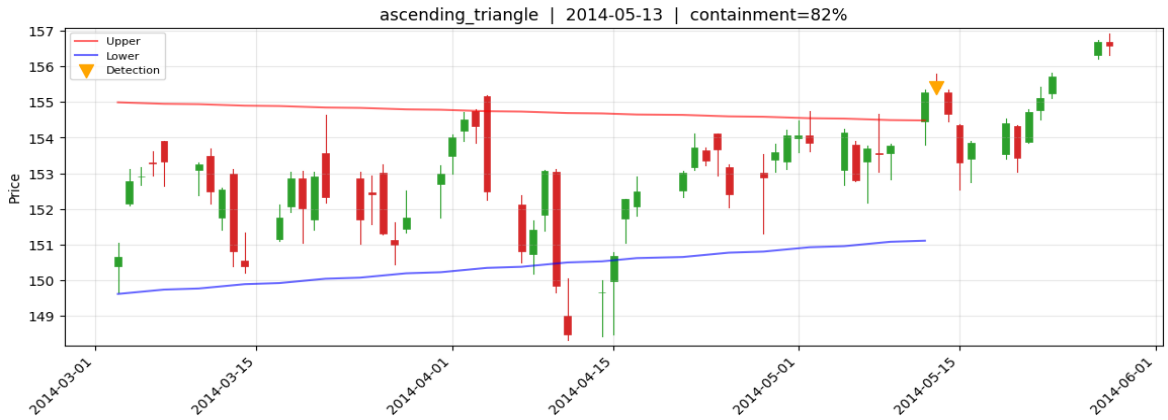
Aspect	Before	After
Data points	All 50 bars' highs/lows	Swing highs + swing lows (≥3 per side)
Regression	np.polyfit (OLS)	scipy.stats.theilslopes (robust median slopes)
Slope thresholds	Absolute (0.01)	ATR-normalised (0.1 × ATR / window)
Containment check	None	≥80% required
Quality metadata	Slope + intercept only	+ containment_ratio, pivot counts

3.3 Channel detector (src/patterns/channels.py)

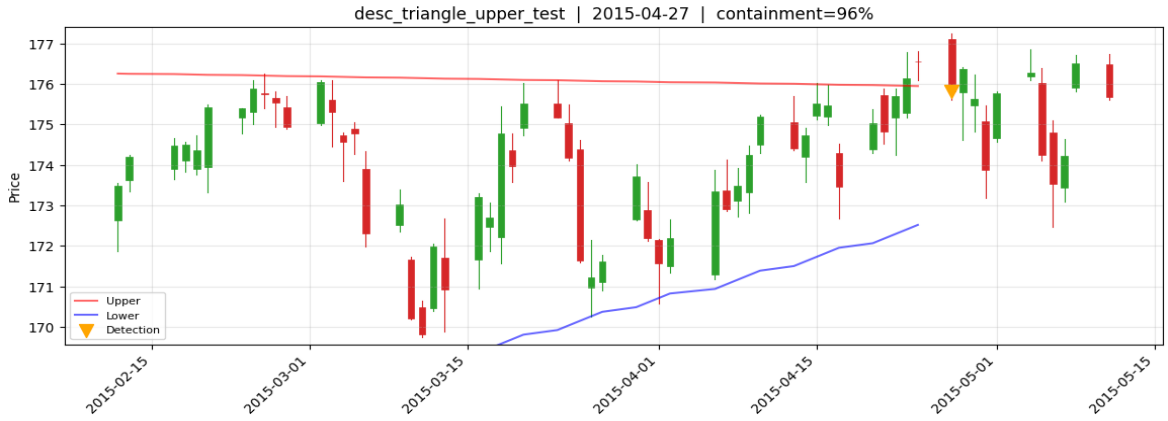
Aspect	Before	After
Data points	All 50 bars' highs/lows	Swing highs + swing lows (≥3 per side)
Regression	np.polyfit (OLS)	scipy.stats.theilslopes (robust median slopes)
R^2 check	On all 50 bars	On pivot points only (more meaningful)
Touch count	Band = 0.3×ATR, ≥4 distinct	Replaced by pivot count (≥3 per side)
Containment check	None	≥70% required

4. Detection Charts (After Fix)

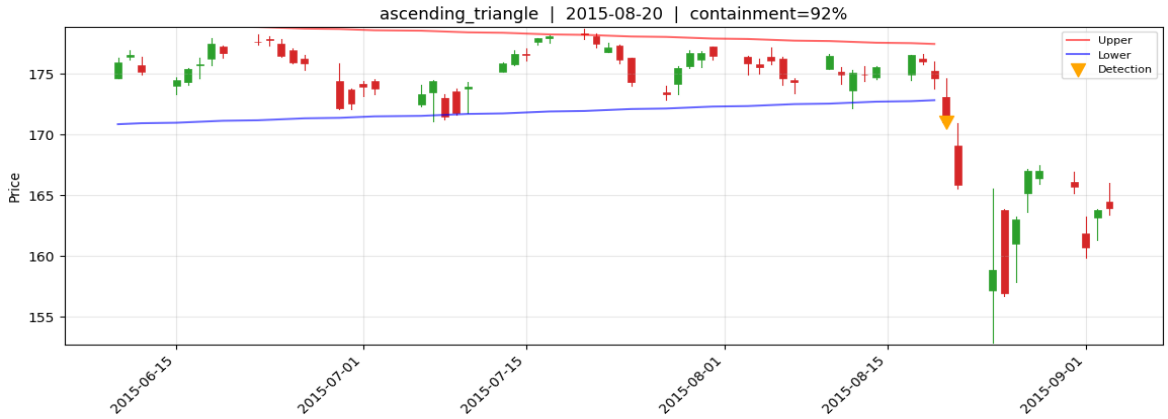
Each chart shows the 50-bar formation window plus 10 forward context bars. Upper trendline (red) fitted to swing highs, lower trendline (blue) fitted to swing lows. Orange marker = detection event.



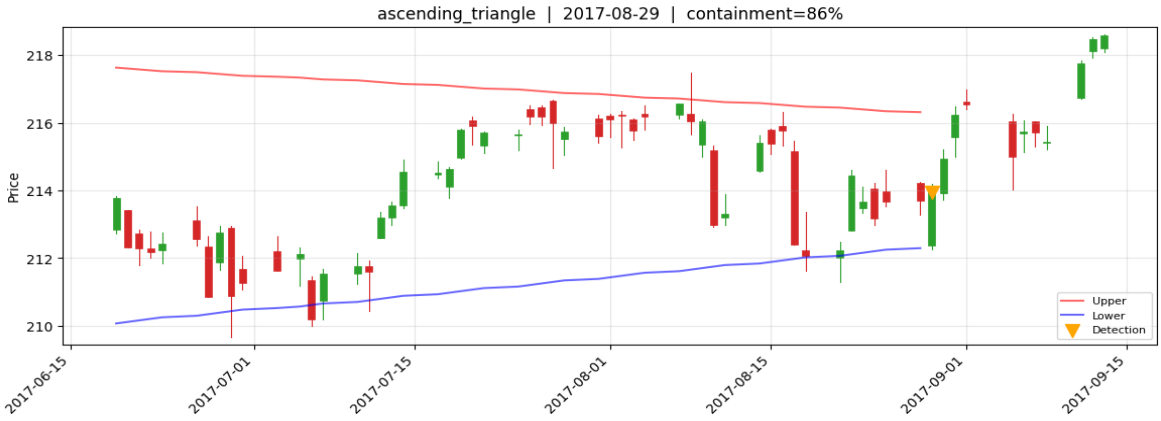
2014-05-13_ascending_triangle



2015-04-27_desc_triangle_upper_test



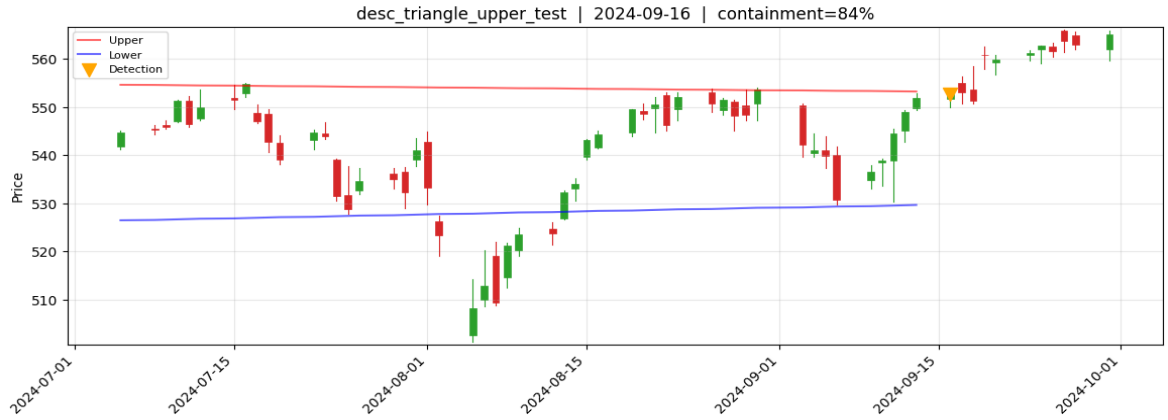
2015-08-20_ascending_triangle



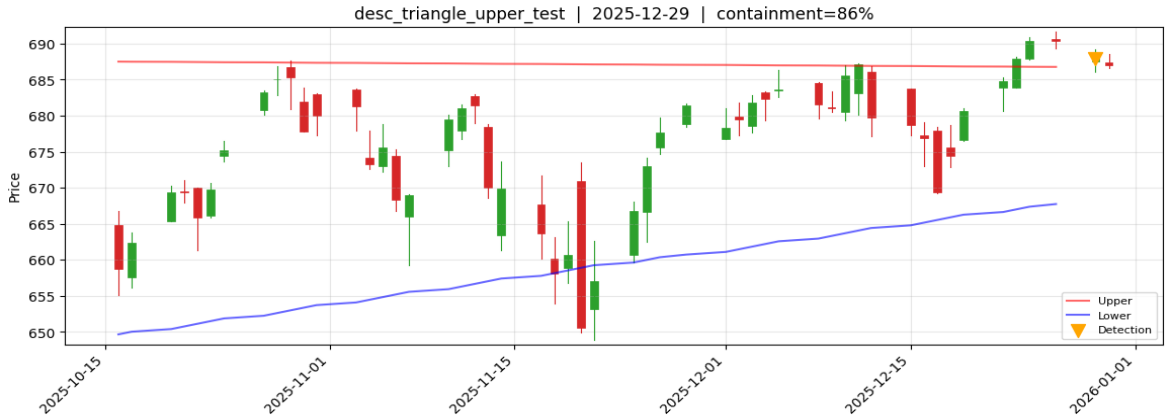
2017-08-29_ascending_triangle

4. Detection Charts (continued)

2022-07-07_channel_down



2024-09-16_desc_triangle_upper_test



2025-12-29_desc_triangle_upper_test

5. Impact on Training Pipeline

Metric	Before fix (tri/ch excluded)	After fix (all patterns included)
Total labeled events	104	111
S/R events	41	41
Multi top/bottom	63	63
Triangles	0 (excluded)	6
Channels	0 (excluded)	1
Long labels	46	47
Short labels	32	36
No-trade labels	26	28

With the fixed detectors, triangle and channel events can be safely re-included in the RF training set. The 7 additional events have verified geometric quality (containment ≥70%) and add pattern diversity to the model.

Recommendation: re-include fixed triangle/channel events in training. Use `label_events(df)` without `exclude_patterns` to get all 111 events.

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6. Conclusion

The pivot-point approach fundamentally solves both issues identified by the supervisor:

- **Triangles:** candles are now contained within the trendlines (82–96% containment vs. ~4% before). Lines are fitted to confirmed swing points, not noise.
- **Channels:** trendlines follow the price envelope without breakpoints (74% containment vs. ~9% before). The strict quality gate means only geometrically valid channels pass.
- **Trade-off:** fewer detections (81 → 7), but every surviving detection is visually correct and structurally sound.

The detection quality is now suitable for ML training and thesis presentation. The containment_ratio metric is also available as a feature for downstream models.